

Automated 3D Stenosis Quantification and Visualization Workflow for Coronary Artery Disease Clinical Decision Support

Adrià Cortés Cugat

UPF Tutor: Pr. Oscar Camara Rey

Supervisor: César Acebes Pinilla

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**Universitat
Pompeu Fabra**
Barcelona



SANT PAU
Campus Salut
Barcelona



Hospital de
la Santa Creu i
Sant Pau

Summary

Coronary Artery Disease (CAD) remains a leading cause of mortality worldwide, placing a significant burden on global healthcare systems. Developed in direct collaboration with Hospital de la Santa Creu i Santa Pau, this project addresses the limitations of current clinical workflows for CAD assessment, which often rely on manual, visual, and subjective quantification of medical images. This traditional approach leads to significant inter-observer variability and requires substantial assessment times for clinicians. This project proposes an end-to-end automated pipeline designed to transform conventional analysis into an objective, geometry-aware clinical decision support system, tailored to the hospital's diagnostic requirements.

The proposed workflow is structured into four integrated blocks. Initially, it performs automated geometry extraction from clinical imaging to generate high-fidelity 3D mesh surfaces and vascular centerlines. Subsequently, a robust stenosis quantification method based on 3D cross-sectional area analysis is implemented. Unlike traditional diameter-based metrics, this area-centric approach correlates more accurately with physiological pressure drops, especially in eccentric plaques. To ensure anatomical fidelity, the methodology incorporates geodesic distance metrics to navigate tortuous arterial paths accurately.

The final stages focus on clinical utility by automating standardized CAD-RADS categorization based on lesion severity. These results are integrated into an interactive visualization dashboard designed to streamline patient prioritization and triage within the hospital setting. By replacing expert-dependent manual measurements with a standardized, automated framework, this tool aims to enhance diagnostic precision and optimize the management of high-risk cardiac patients at Hospital de Sant Pau, ultimately facilitating faster and more objective clinical interventions.